DAY 17 OF 30

Event-stream

Catching data as it happens.
And shaping it before it lands.

Sources **Transform** **Route**

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THE PROBLEM

Events come from everywhere, in every shape.

1	**Apps**	Clicks and orders
2	**Devices**	Sensor readings
3	**Databases**	Row changes
4	**Other clouds**	Kafka and message queues

Each one used to need **its own custom code to collect, clean and deliver**.

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THE SIMPLE VERSION

A conveyor belt with sorting stations.

Parcels drop onto a belt from many loading doors. Along the way they are scanned, sorted and sent down the right chute.

An eventstream is that belt: **events in, shaped in motion, delivered where they are needed.**

✓ **Sources**	Where events come from
✓ **Operators**	What happens on the way
✓ **Destinations**	Where they end up

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HOW IT WORKS

Building an eventstream

1. **Add sources**: Event Hubs, IoT Hub, Kafka, database changes
2. **Shape** on a canvas: filter, aggregate, join, group by
3. **Route** to an Eventhouse, a Lakehouse or Activator
4. **One stream** can feed several destinations at once

No code required

The whole flow is drawn on a canvas. A SQL operator is there for logic that needs it.

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FOR ENGINEERS

What engineers should know

It speaks Kafka

An eventstream exposes a Kafka-compatible endpoint, so existing producers can send to it with little change.

Database changes are a source

Change data capture from SQL Server, PostgreSQL, MySQL, Cosmos DB and others can flow in as events.

Delivery is at-least-once

An event can arrive twice. Downstream logic should tolerate duplicates.

Windows are how you aggregate

Group by time windows to turn an endless stream into counts and averages.

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THE HONEST PART

Eventstream or a pipeline?

EVENTSTREAM	PIPELINE
Data arrives continuously	Files arrive on a schedule
You need it within seconds	Hourly or daily freshness is fine
Several consumers need the same events	The source can only be queried in bulk

> **The catch:** filtering late is expensive. Every event you carry to the destination costs storage and compute. Drop what you do not need as early as possible.

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FROM REAL WORK

Eventstream on a real project

A consumer delivery platform with streaming operational data.

1. **Event Hub** received the raw events at high velocity
2. **Eventstream** carried them into Fabric
3. **Eventhouse** stored them for KQL queries
4. **Leadership** asked for live numbers in natural language

Terabytes per minute, with **sub-second answers** at the end.

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SAVE THIS

Eventstream in five lines

1. **Many sources**, one way to connect them
2. **Shape events** while they move
3. **Send them** to more than one place
4. **Expect duplicates**, design for them
5. **Filter early**, pay less later

Tomorrow: Eventhouse and KQL. Asking questions of fast-moving data.

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